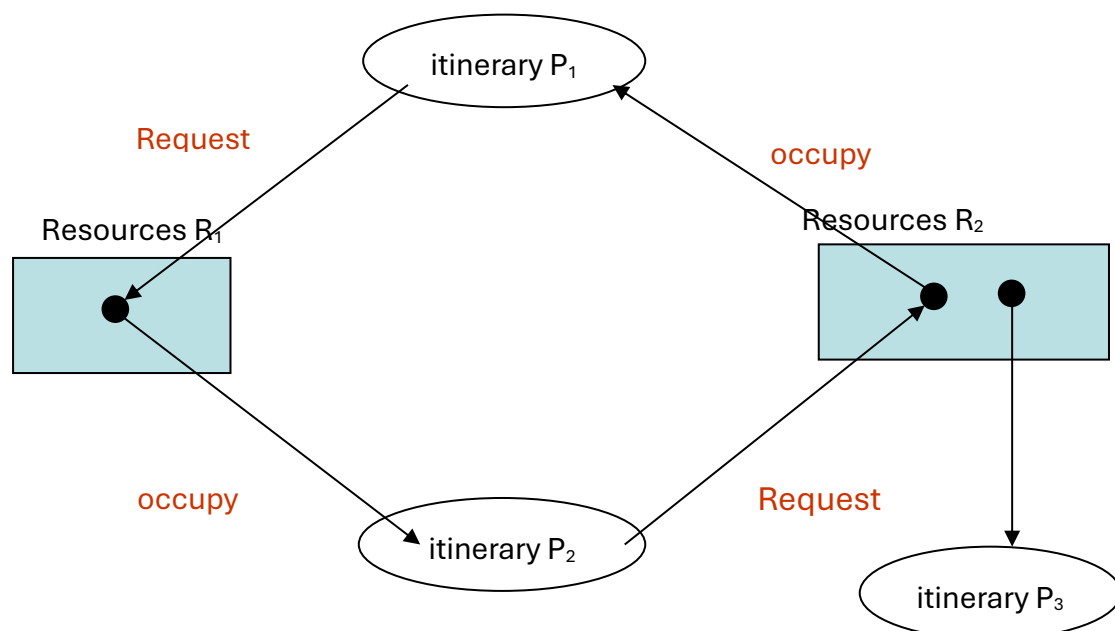


Chapter 3

A. Questions

1. Please explain the differences between independent and cooperative itineraries.
2. Why does a collaborative itinerary need a synchronization mechanism?
3. Explain what is a competition situation. How can we avoid competition situations?
4. Explain what are the critical zone, entry zone, and exit zone.
5. Explain and explain the three conditions that satisfy the critical zone.
6. In the philosopher's dinner question, besides the methods listed in this section, explain what other methods can prevent deadlocks.
7. What is a deadlock? Please explain the four conditions for a deadlock to be established.
8. Please explain how you can break the deadlock? What factors need to be considered?
9. Suppose there is a closed loop in the resource allocation diagram. Is it inevitable that a dead end will occur?

For example, in the diagram below :



10. Suppose a system has a total of 8 R. Please determine whether the following systems are in a safe state. If so, write a secure execution sequence for it.

itinerary	The total number of R is required	The number of R held
P1	8	1
P2	5	2
P3	1	1
P4	7	2
P5	3	0